

Module 14 Final

535.632 Applied Finite Element Analysis

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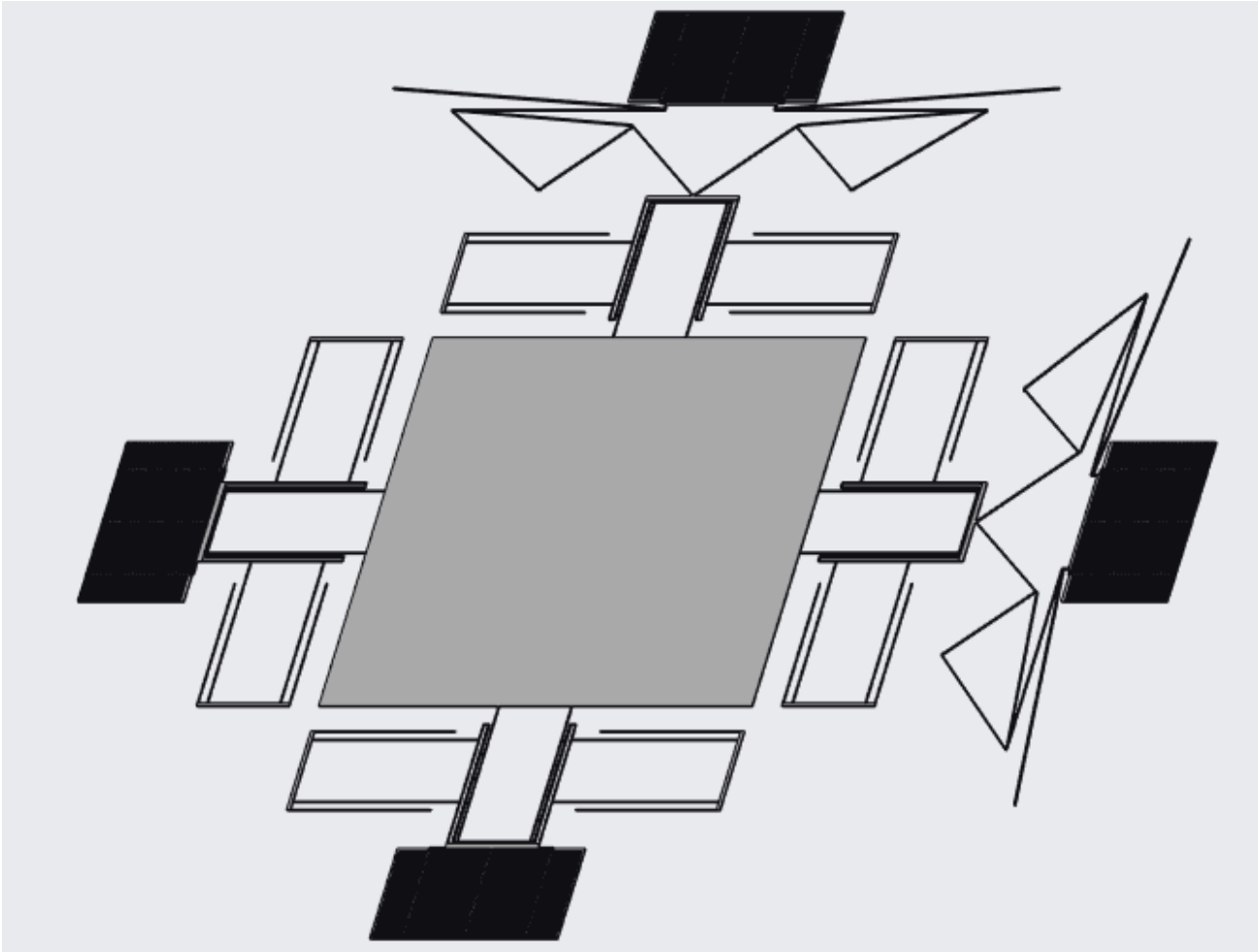
Accelerometer

This project involves the finite element analysis (FEA) of a dual axis, capacitance based accelerometer. The goal is to assess the structural and thermal performance under various load conditions, namely:

- Degenerate Mode Shapes for the ideal operation of the accelerometer
- Finding the harmonic response behavior of the accelerometer
- Response of the accelerometer under 1G of loading
- Verification of the Displacement Amplifying Compliant Mechanism
- Understanding the heat distribution of due to the Capacitance nature of the accelerometer, and where does it get concentrated.

Geometry and Materials

- Geometry: The accelerometer is similar in both axis, hence simulation can be done in only one axis to get the full axis parameters of the MEMS device. It also contains 4 comb drives, which are excluded for this particular simulation.
- Materials Assignments:
 - Accelerometer: Pure Silicon



Overview of load cases

- Modal Analysis
- Harmonic Response
- Acceleration Under 1G
- Acceleration under 1G with Varying Temperature
- Heating due to capacitance

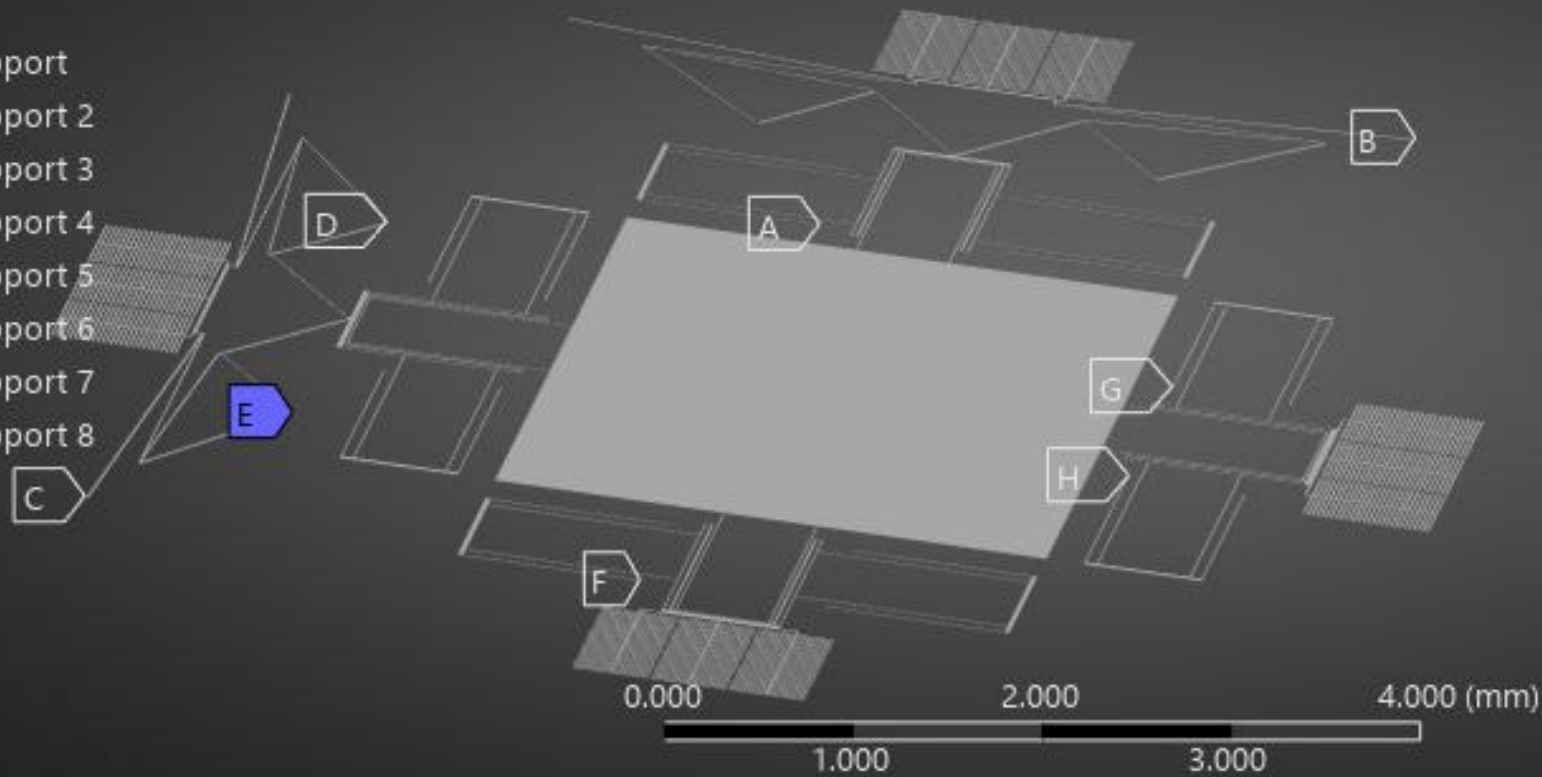
Modeling assumptions and challenges

- One end of the model was assigned zero displacement constraint at fixture points and no force transfer or heat transfer.

Fixed Constraints

A: Modal
Modal
Frequency: N/A
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- A** Fixed Support
- B** Fixed Support 2
- C** Fixed Support 3
- D** Fixed Support 4
- E** Fixed Support 5
- F** Fixed Support 6
- G** Fixed Support 7
- H** Fixed Support 8



Meshing

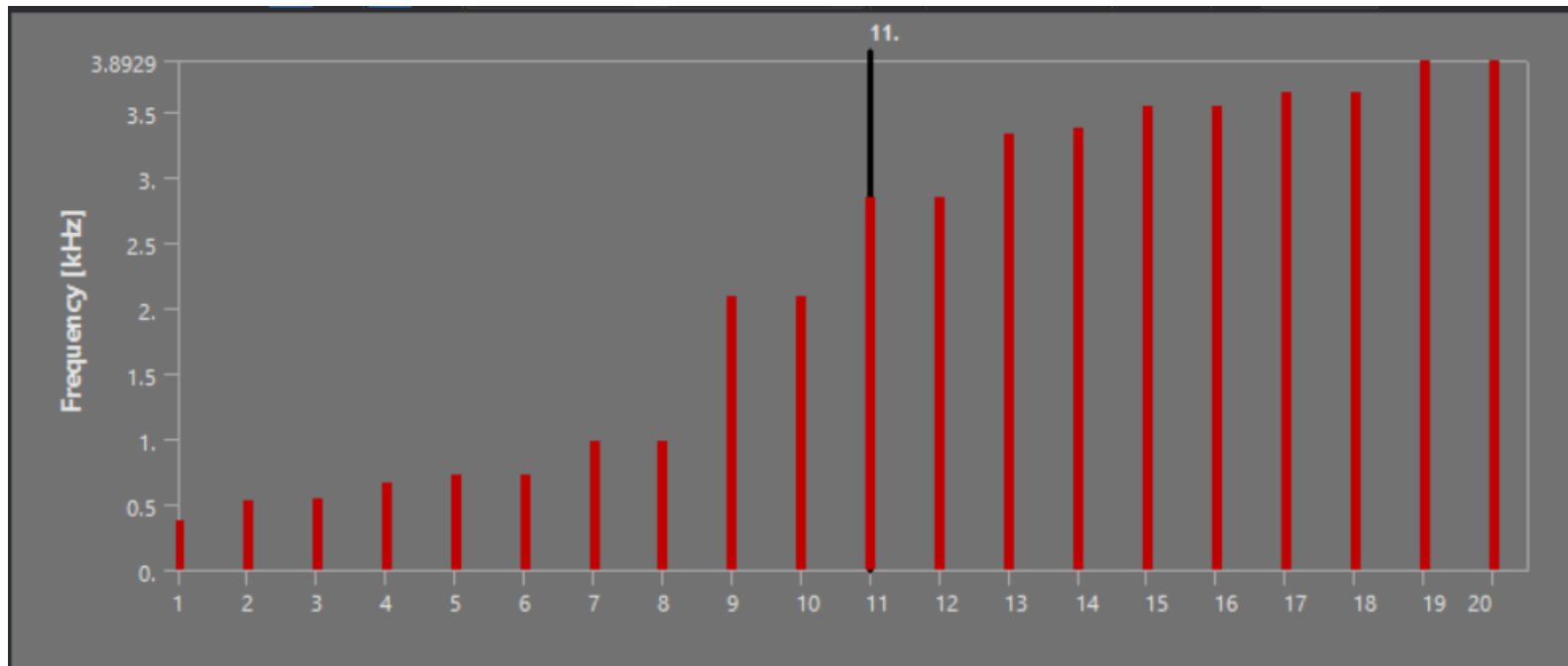
- Originally started with a mesh control, tried multiple maximum element
- Was unable to hex mesh due to the fact that most of the beams were slender.
- Hence decided to use the auto-generate mesh to create the results. Improved the fineness of the mesh quality to reach appropriate confidence in the results.

Modal Analysis

Expected Results of Modal Analysis

Find the first 20 modes of the accelerometer. Find the 2 degenerate modes for the optimal operation of the accelerometer in the active mode.

The degenerate mode shapes of interest were Mode 11 and 12 at 2.84 kHz

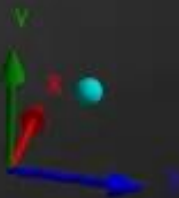
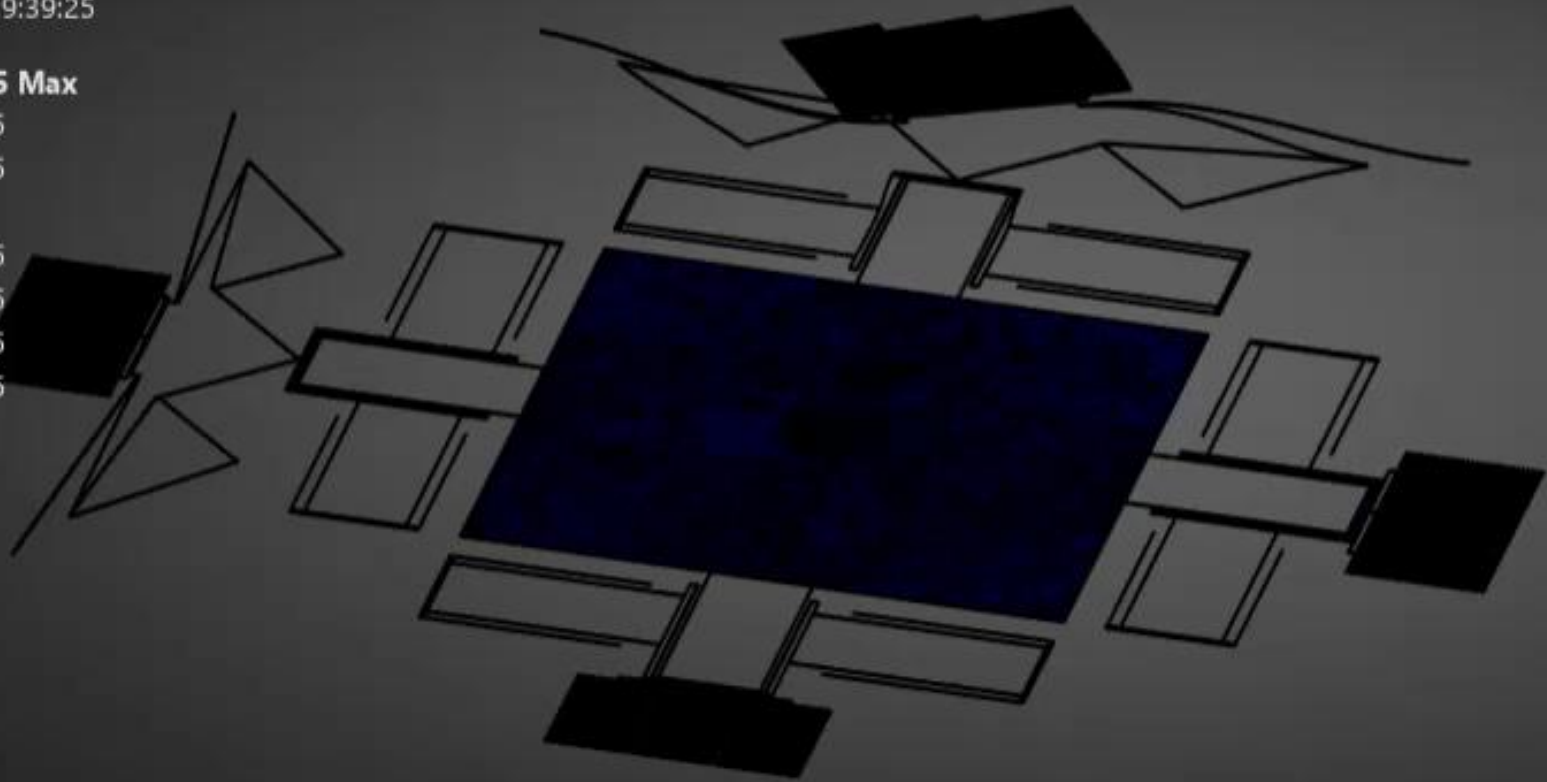
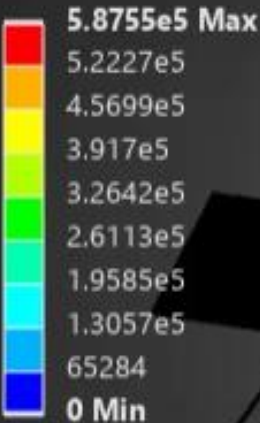


	Mode	☒ Frequency [kHz]
1	1.	0.37142
2	2.	0.52488
3	3.	0.53922
4	4.	0.66124
5	5.	0.72986
6	6.	0.73044
7	7.	0.9901
8	8.	0.99041
9	9.	2.0951
10	10.	2.0963
11	11.	2.8533
12	12.	2.854
13	13.	3.3291
14	14.	3.38
15	15.	3.5413
16	16.	3.5426
17	17.	3.6474
18	18.	3.6488
19	19.	3.892
20	20.	3.8929

Mode 11

A: Modal
Total Deformation
Type: Total Deformation
Frequency: 2.8533 kHz
Unit: mm
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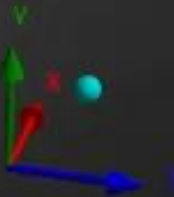
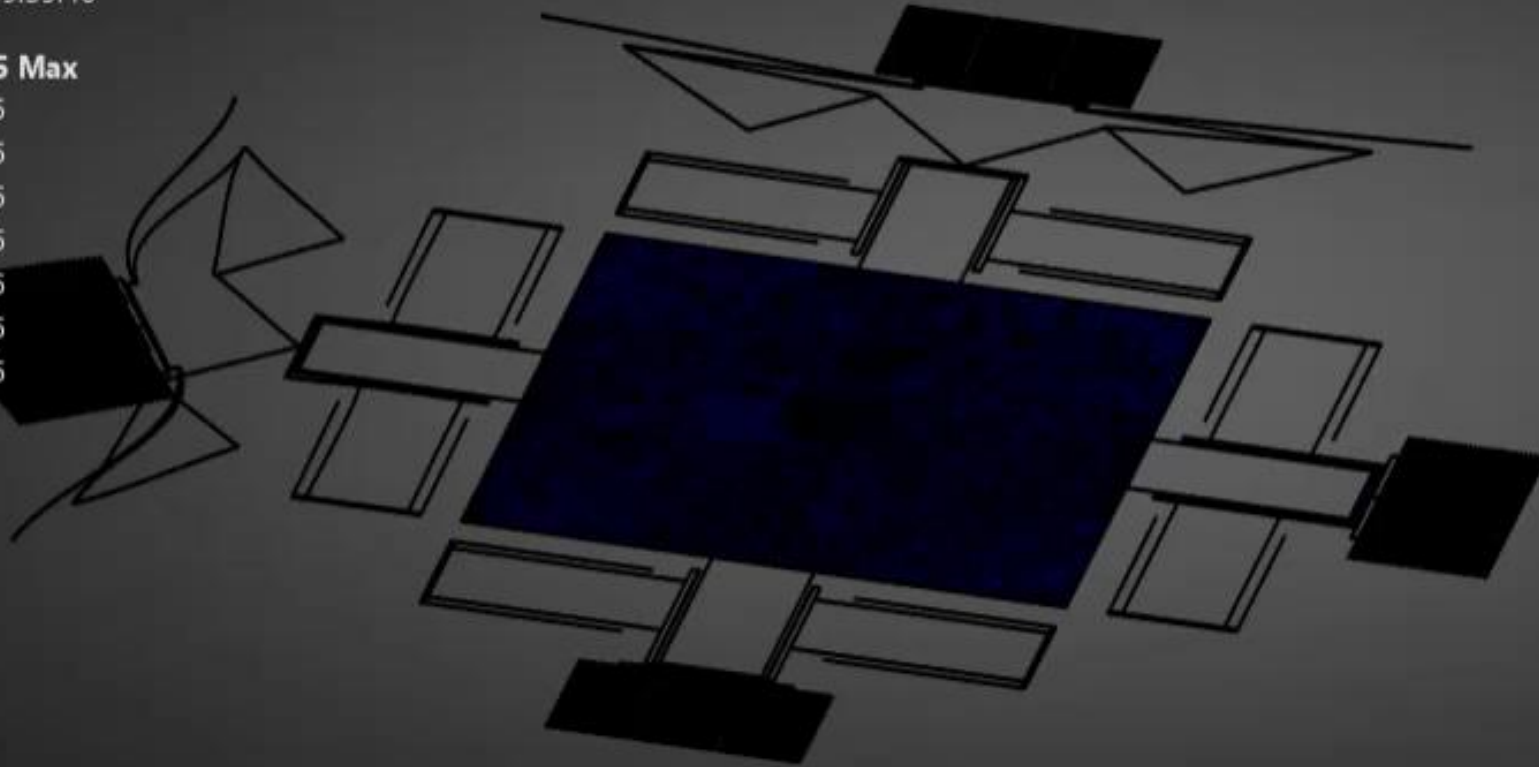
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Mode 12

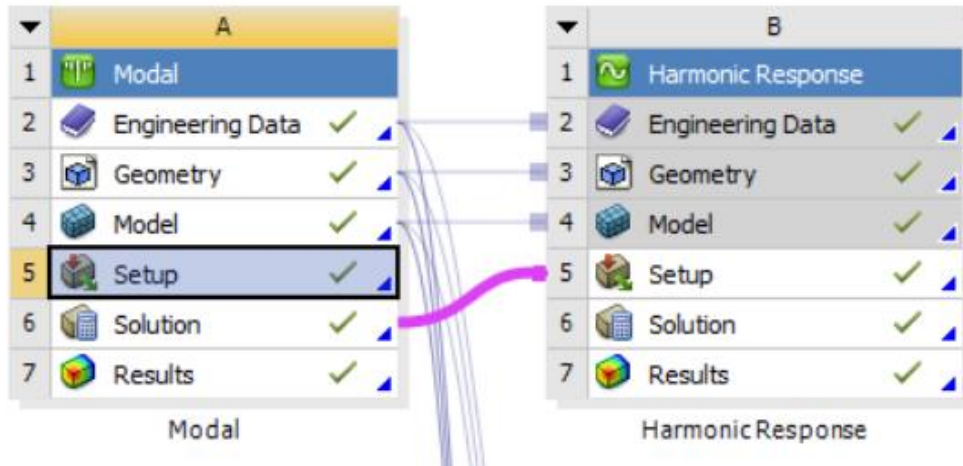
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A: Modal
Total Deformation 2
Type: Total Deformation
Frequency: 2.854 kHz
Unit: mm
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Harmonic Response under 1G
acceleration at operational
Frequency (2.84 kHz)

Setup



The modal frequency results are fed into the Harmonic Response system. This would sweep between the defined frequency and the response of the device under 1G.

We could also add the damping coefficient of the accelerometer, but due to the lack of time couldn't explore it.

Step Controls	
Multiple Steps	No
Options	
Frequency Spacing	Linear
Range Minimum	2. kHz
Range Maximum	3.2 kHz
Solution Intervals	10
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
On Demand Expansion Option	Program Controlled
-- On Demand Expansion	No
Store Results At All Frequencies	Yes
Mode Selection Method	None
Rotordynamics Controls	
Output Controls	
Damping Controls	
Analysis Data Management	

Harmonic Response at 1G at 2.84 kHz

B: Harmonic Response

Total Deformation

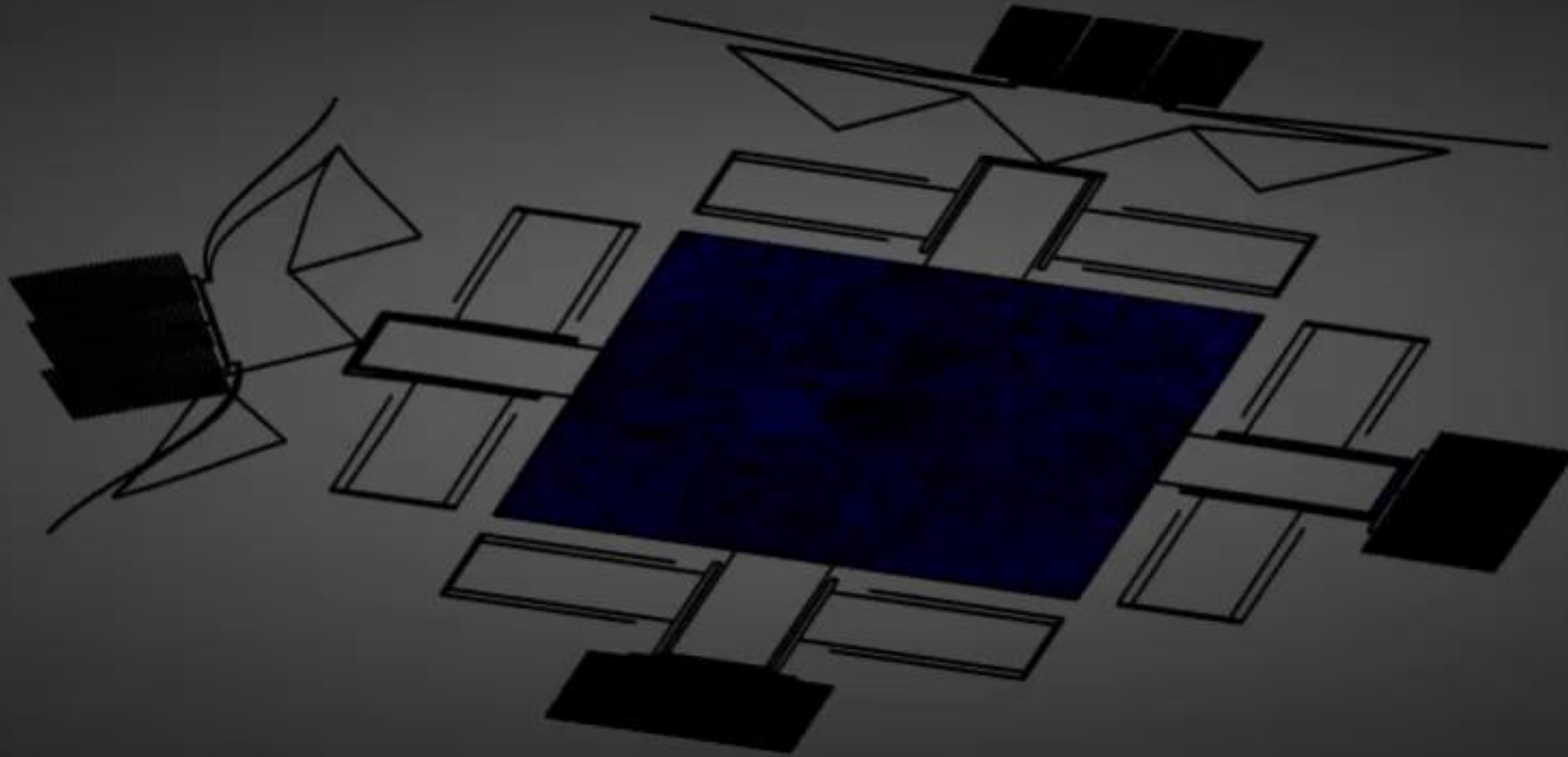
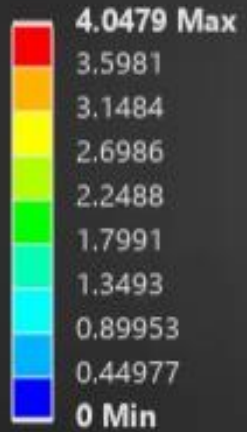
Type: Total Deformation

Frequency: 2.84 kHz

Sweeping Phase: 0. °

Unit: mm

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Results

- The harmonic response of the device, proved that the device is working accurately for the following reasons:
 - Only one of the comb is deforming more than others, as per the design. The acceleration is applied only in 1 direction.
 - The comb that is deforming the most is due to the displacement amplifying compliant mechanism; hence the opposite side should have minimal to no deformation compared to the other comb.

Acceleration at 1G

Static Structural Study

Loading Conditions

C: Static Structural

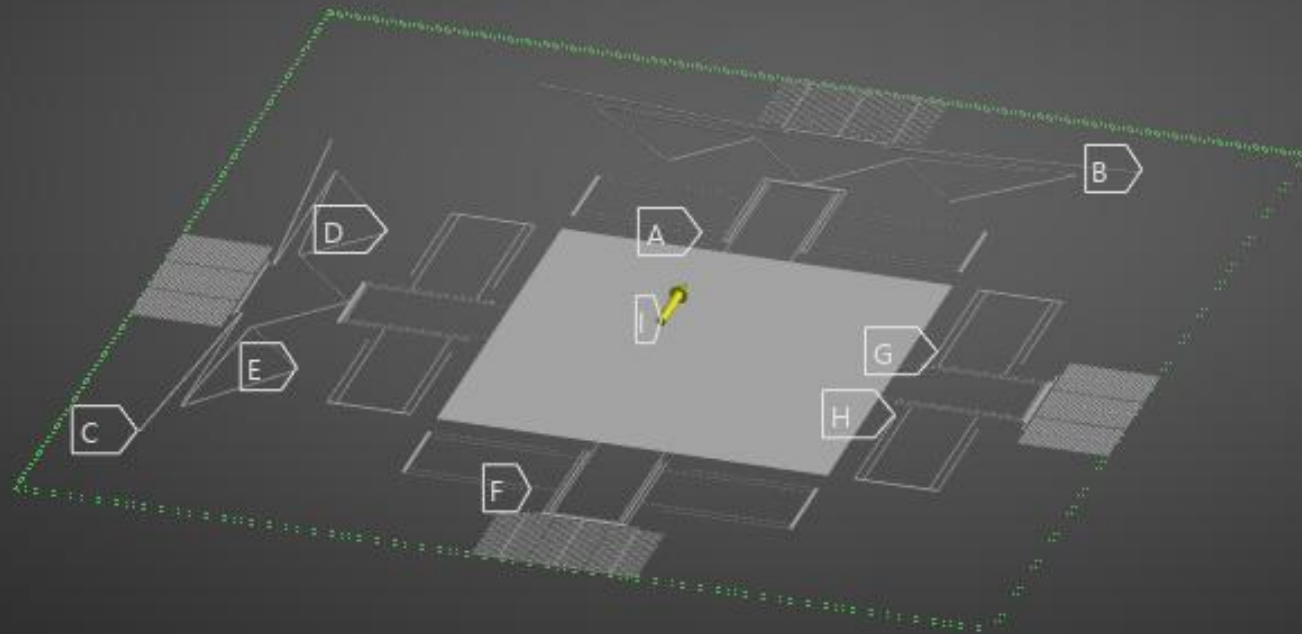
Static Structural

Time: 1000. ms

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- A** Fixed Support
- B** Fixed Support 2
- C** Fixed Support 3
- D** Fixed Support 4
- E** Fixed Support 5
- F** Fixed Support 6
- G** Fixed Support 7
- H** Fixed Support 8
- I** Acceleration: 98. mm/ms²



0.000 1.250 2.500 3.750 5.000 (mm)

Displacement

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C: Static Structural

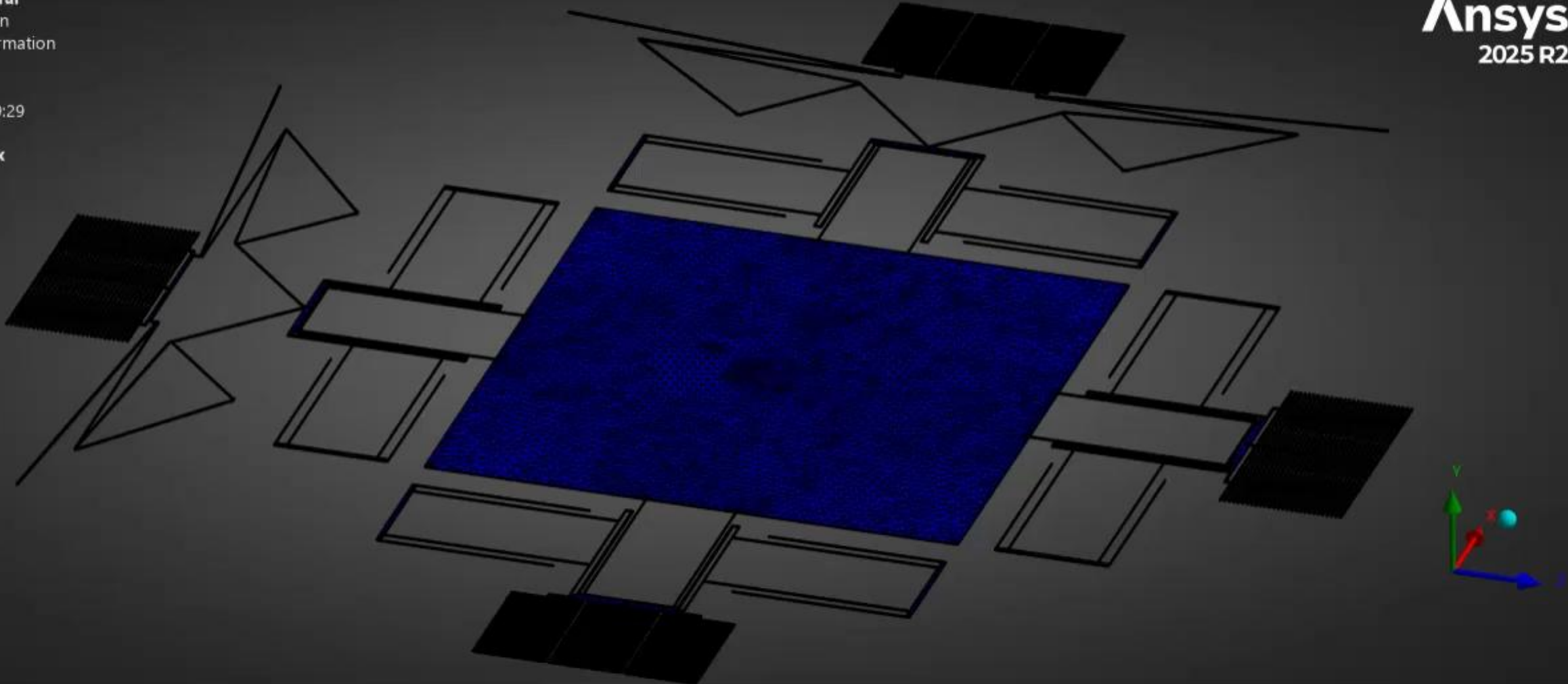
Total Deformation

Type: Total Deformation

Unit: mm

Time: 0 ms

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Displacement

C: Static Structural

Total Deformation

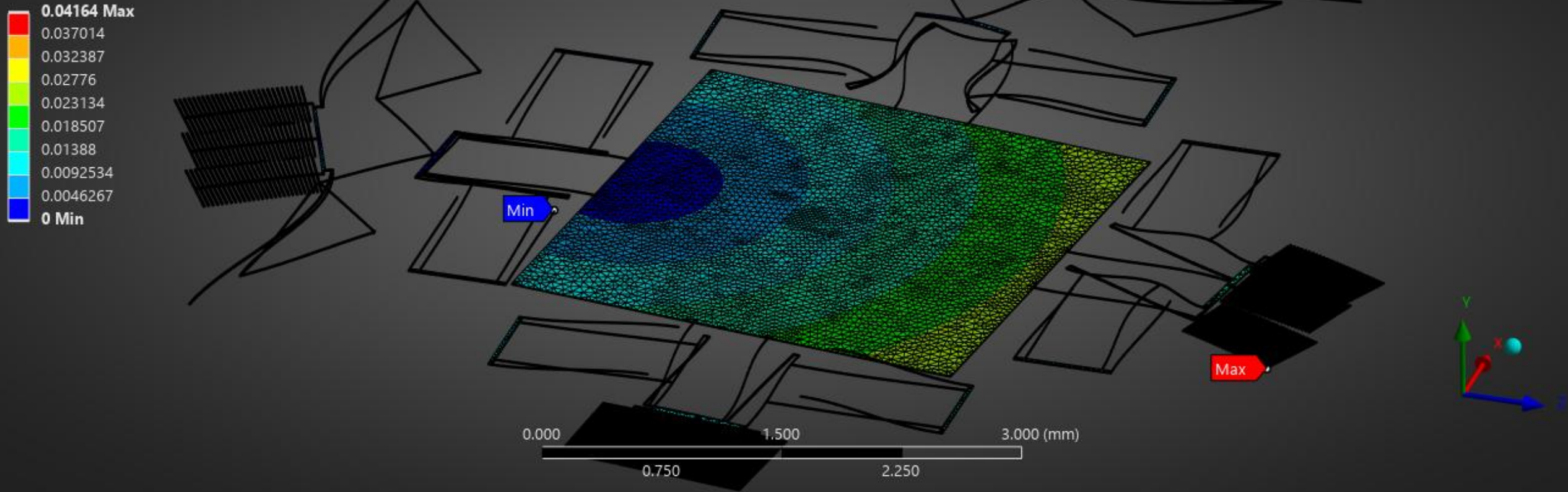
Type: Total Deformation

Unit: mm

Time: 1000 ms

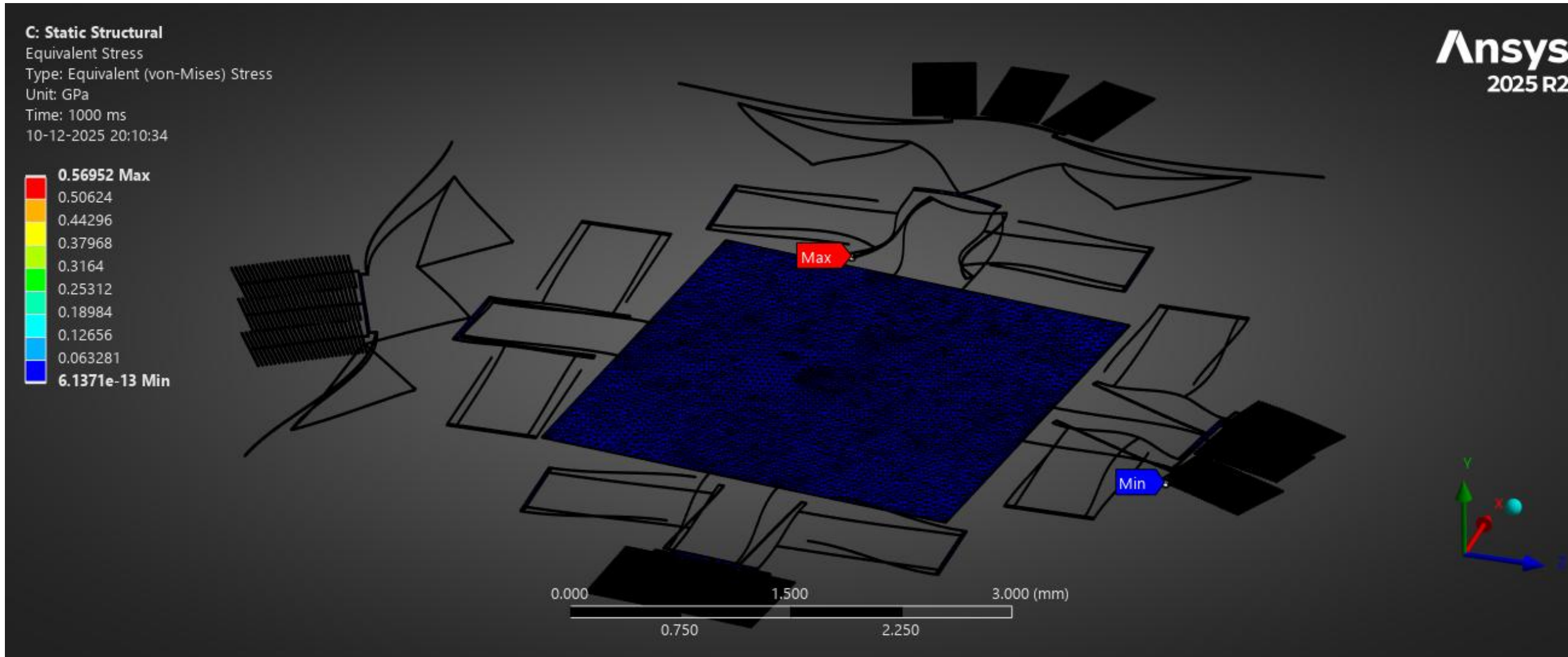
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Acceleration Applied in the X direction

Stress (Von Mises)



Acceleration Applied in the X direction

No Major stress regions that would indicate failure, hence device is under the required stress limits

Results

Out of Plane Thickness in Micron		10
Eigenfrequency		2867.7
Displacement Sensitivity($\mu\text{m/g}$)		3.7
Capacitance Sensitivity for Full Comb Drive($\Delta C/g$)		1.66
$\Delta C/C_0$ for μg		7.79E-07
Cross-Axis Sensitivity (nm/g)		3.2
Mass of the Accelerometer		2.39E-07
Effective Mass of the Accelerometer		8.51E-09
Quality Factor- Analytical		4.92E+02
Brownian Noise Equivalent Acceleration		6.31E-07

Equation

$$C = \frac{\epsilon_0 A}{d}$$

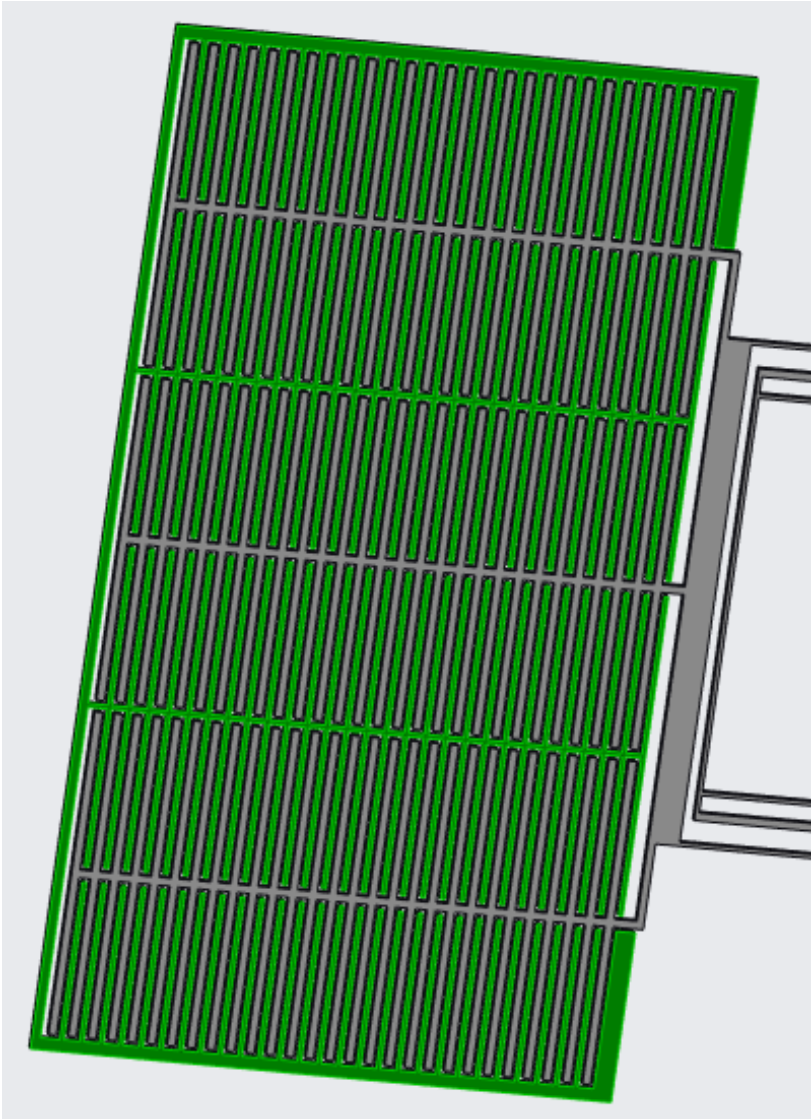
Diagram illustrating the equation for Capacitance (C):

- ϵ_0 : Electrostatic constant
- A : Area
- d : Distance Between plates

- We were able to extract the performance characteristics of the accelerometer. This would allow us to compare with various accelerometers present in the market.
- We could improve the accelerometer by actually reducing the sensitivity to a reasonable limit to prevent actuation due to noise. This is defined by the quality factor and the Brownian Noise Equivalent acceleration.
- For the capacitance sensitivity, the area of the comb and using the above formulation.

Heating Effects of Electrostatic Drives

Problem Statement



Due to the electrostatic actuation, between the Comb drive and the Accelerometer, we notice heating.

Through steady state thermal Analysis we can identify the regions in which the highest amount of heat generation occurs.

We consider a heating flow of 10kW on the comb's surfaces, while this doesn't translate to the actual value, it gives us an idea of the localization of the heat in the accelerometer. It would let us create any necessary design changes to counteract the same.

Setup

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D: Steady-State Thermal

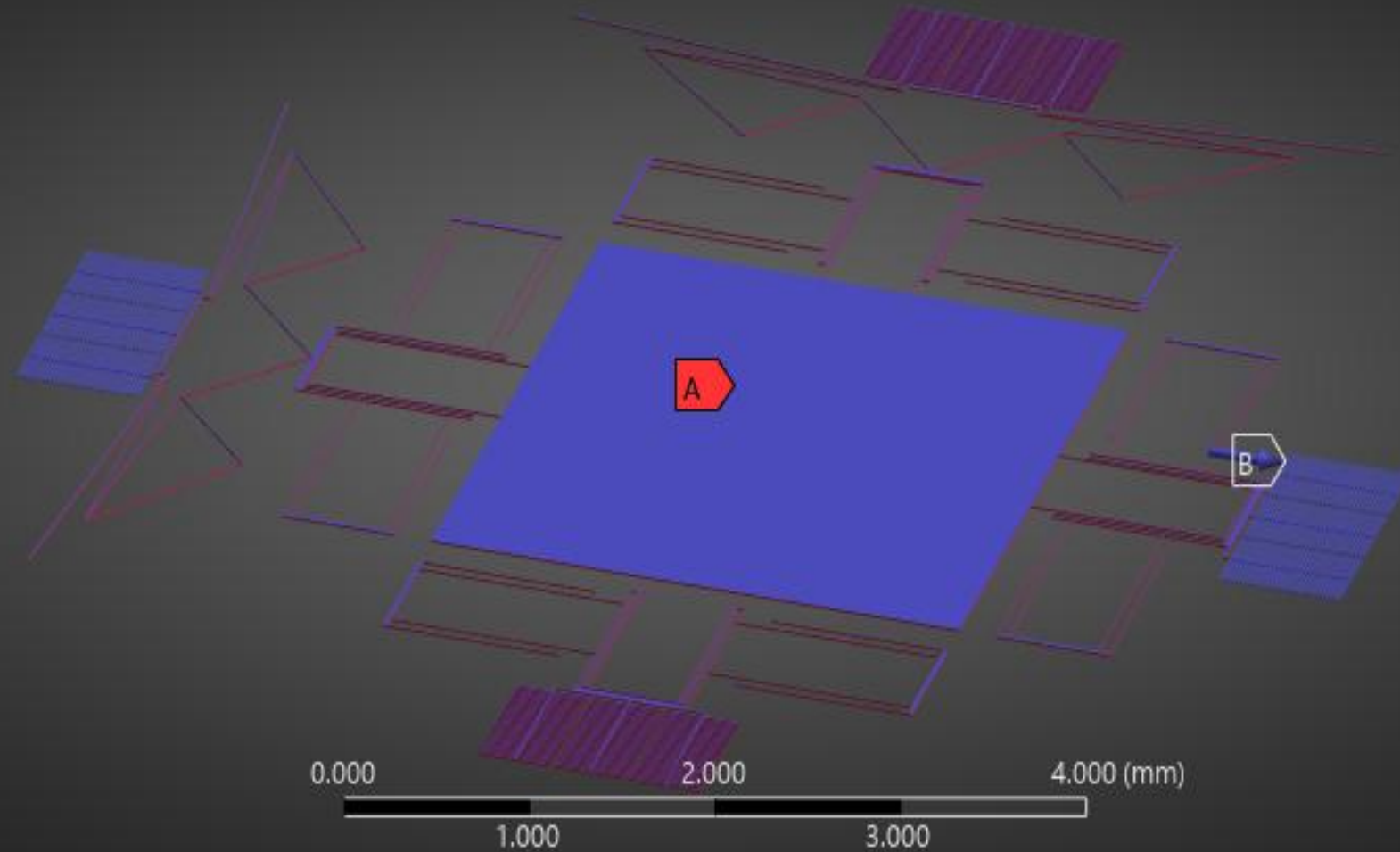
Steady-State Thermal

Time: 1000. ms

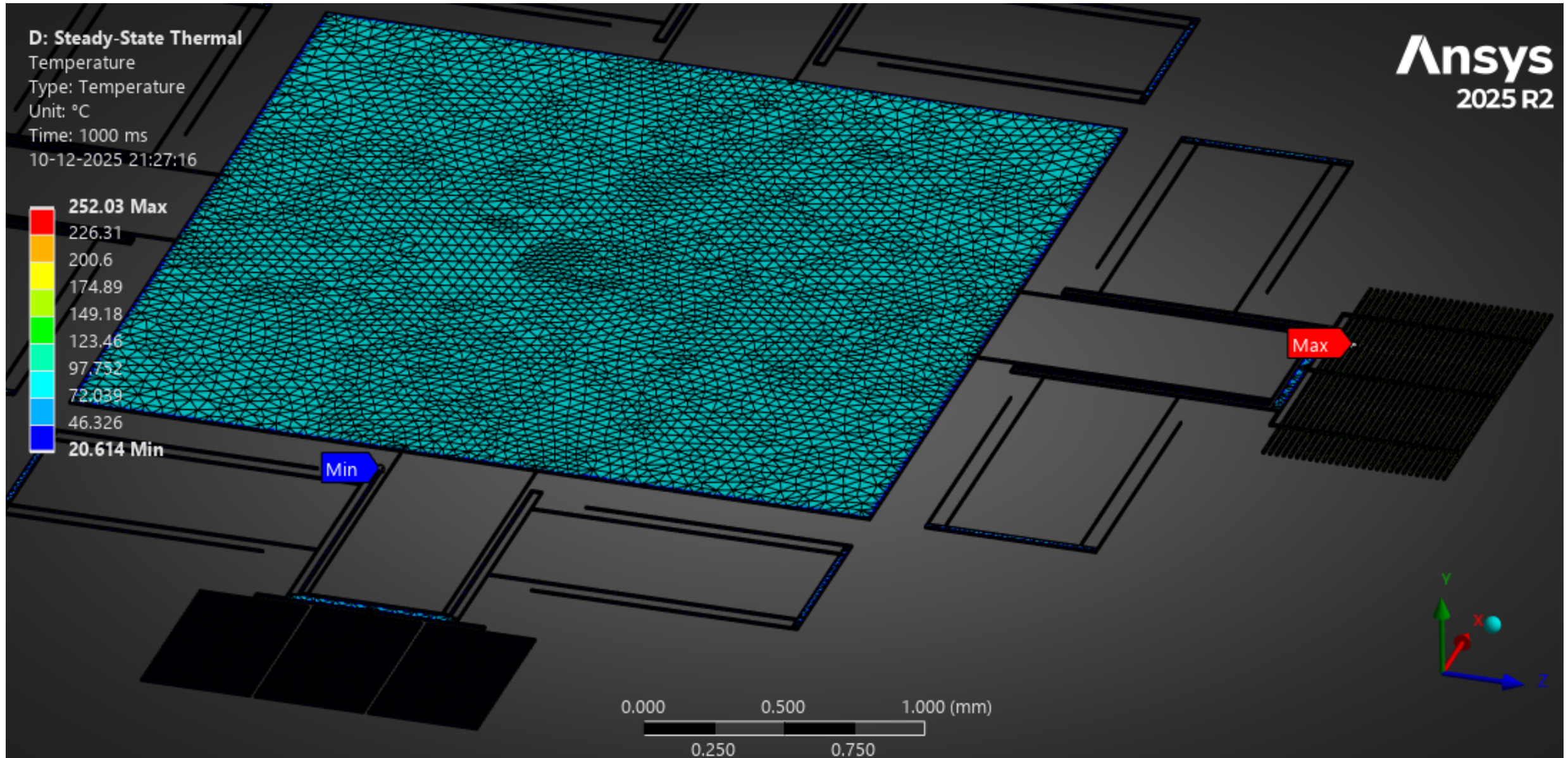
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A Temperature: 22. °C

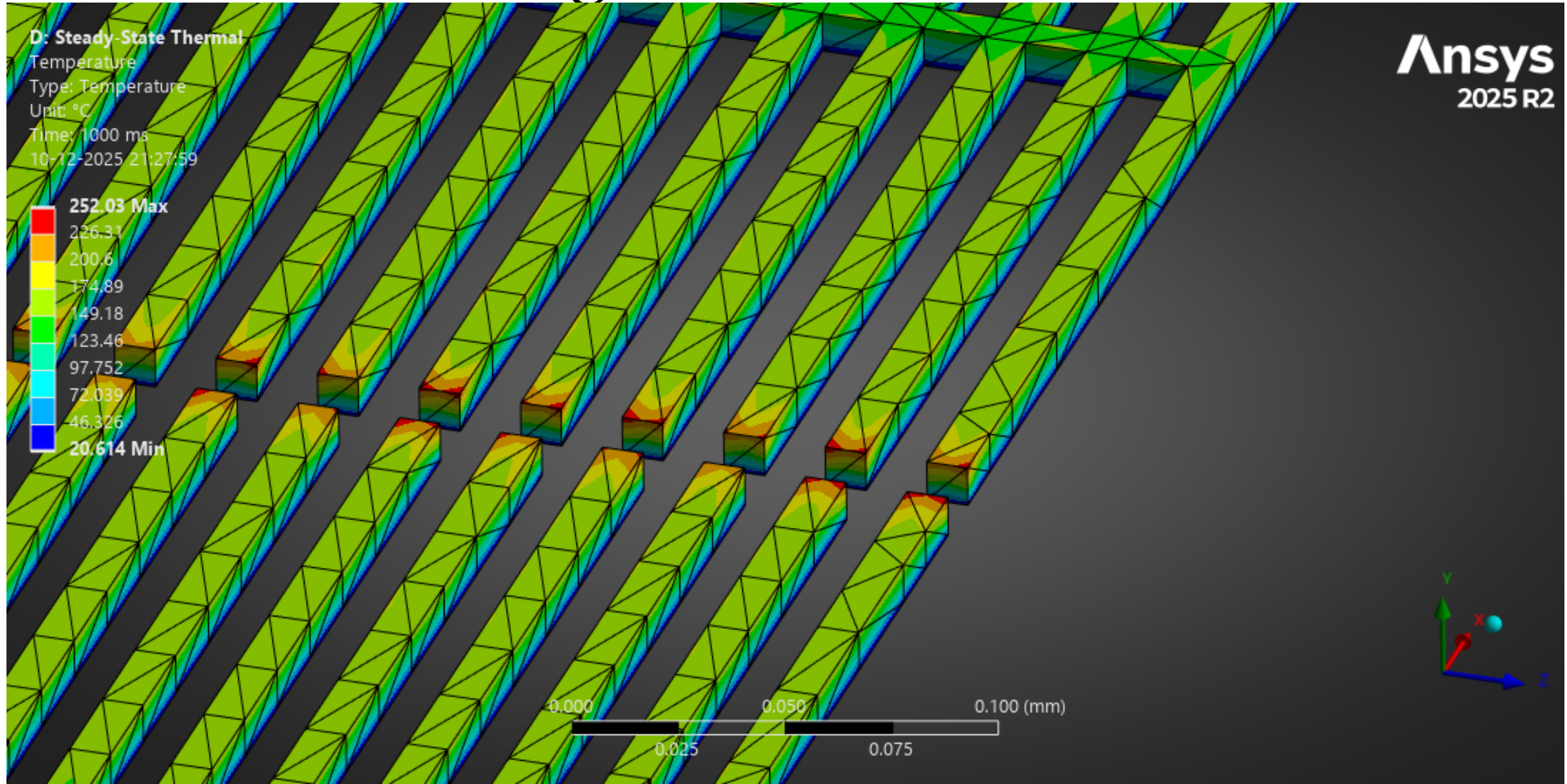
B Heat Flow: 10. kW



Results



Maximum Heating effect on the comb drive



Results

- The overall heat generated is negligible, and the maximum heating occurs on the ends of the comb drive. Hence this doesn't majorly affect the performance of the accelerometer.

Conclusion

- Designed a 2-axis accelerometer matching the design specification and is able to handle the required loading conditions.
- Has 2 degenerate mode shapes to be actively actuated during the operation of the accelerometer.
- Harmonic performance of the accelerometer at 2.84 kHz has shown to be close to reality.
- Heating effects due to electrostatic interaction is negligible and doesn't affect the working of the accelerometer.

Thank you